MOHAMED EL BAAMRANI

Full-Stack & AI Software Engineer

+212 7 08 82 52 53 · medbaamrani1@gmail.com · Paris, France · github.com/medbm5

SUMMARY

Software engineer with 3+ years of production experience, from Python/Node.js backends to React/TypeScript front ends, on a healthcare platform where data reliability and traceability are critical. Integrated 6+ clinical systems (Epic, EMIS, SystmOne, ADT feeds) via FHIR, HL7, REST, SOAP and Mirth Connect, and built the backend of a GenAI platform at Allianz Global Investors (RAG, Azure OpenAI, Semantic Kernel), taken from MVP to production budget approval. Ships complete products end to end, from a trilingual fintech PWA to LLM applications with guardrails. MSc Machine Learning and 5 AWS certifications; tests, strict typing and CI/CD by default.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, PHP, Java, Kotlin, C#, HTML/CSS

Backend: FastAPI, Flask, Pydantic, Node.js, Express, Laravel, Spring Boot, REST APIs, SOAP, microservices, CQRS, Clean Architecture, webhooks, Server-Sent Events

Frontend & Mobile: React, Next.js (App Router, Server Components), Redux Toolkit, Zustand, TanStack Query, React Hook Form, Zod, Tailwind CSS, shadcn/ui, React Native, Expo, Electron, PWA, i18n (RTL), WCAG 2.2 AA accessibility

AI / GenAI & Data: LLMs, RAG, Azure OpenAI, OpenAI API, Azure AI Search, Semantic Kernel, LangGraph, agents, structured outputs, LLM evaluation, pgvector, pandas, NumPy, Polars, DuckDB, scikit-learn, PyTorch, PySpark, Databricks, Delta Lake, MLflow, ETL pipelines, Monte Carlo simulation, Bayesian networks

Healthcare Interoperability: FHIR R4, HL7 v2 (ADT), SMART on FHIR, Epic, EMIS, SystmOne, Mirth Connect (integration engine)

Databases: PostgreSQL (Row-Level Security, pgTAP), Supabase, SQL Server, MySQL, MongoDB, Neo4j, Redis, SQLite

Cloud & DevOps: AWS (EC2, S3), Azure, Docker, Kubernetes, Terraform (IaC), CI/CD, GitHub Actions, Argo CD, GitOps, Prometheus, Grafana, Datadog, Sentry, Vercel, Fly.io, Linux

Security: OAuth 2.0, PKCE, JWT, RBAC, Row-Level Security, Stripe Identity (KYC), privacy by design, audit logging

Quality & Methods: pytest, Vitest, Jest, React Testing Library, Playwright, mypy, TDD, code review, ADRs, Agile Scrum, Jira, Confluence, Git

PROFESSIONAL EXPERIENCE

Software & Data Integration Engineer — Masarat App (eKare inSight) **Apr 2022 – Sep 2023 · Oct 2024 – Present**

Rabat · AI-based wound measurement and tracking platform (web + mobile), agile team of 8

- Integrated **6+ external healthcare systems** (Epic, EMIS, SystmOne, SAP/OAS ADT feeds) via FHIR, HL7, REST, SOAP and Mirth Connect, where data reliability and traceability are critical.
- Built FHIR integrations filing wound assessments into Epic flowsheets (Observation API) with clinical value mapping and normalization, and delivered integrations writing wound data and PDF reports into SystmOne and EMIS.
- Implemented a SMART on FHIR EHR launch from Epic (OAuth 2.0 with PKCE, id_token validation against JWKS) and migrated Epic FHIR calls from Basic auth to OAuth 2.0 backend services (RS384 JWT client assertion), documented for the client.
- Designed ETL pipelines (Mirth Connect + Python) to import/export patient data, with automated integrity checks catching inconsistencies before production; validated HL7 ADT feeds end to end against the SQL Server database.
- Developed web features end to end with Laravel, Node.js and React/TypeScript, and mobile features with React Native; led an **Angular → React migration that cut page load time by 35%**.
- Administered AWS infrastructure (EC2, S3), Docker-based Mirth Connect deployments and Terraform; ran incident monitoring and root-cause analysis (e.g. a UTC vs. BST timezone bug), and introduced change tracking across SIM, Staging and Production after a post-incident review.

Freelance Full-Stack Developer — NUR (client project, Quebec) **2026**

Trilingual (FR/EN/AR) PWA connecting students with mosques offering interest-free loans · Next.js 15, Supabase, Stripe

- Built a three-sided platform (students, mosques, super admin) end to end: tiered loans, multi-loan funding plans and automated monthly repayments.
- Built payments on **Stripe Connect Express** with direct charges so repayments go straight to each lender and the platform never holds funds, plus Checkout, Billing and Identity (KYC), driven by idempotent, signature-verified webhooks.
- Enforced security in the database: PostgreSQL Row-Level Security on every table, loan state transitions through SECURITY DEFINER functions with an audit trail, pgTAP tests proving tenant isolation; privacy by design with 60-second signed URLs.

- Shipped 3 languages including right-to-left Arabic, targeting WCAG 2.2 AA; quality pipeline with strict TypeScript, Vitest, Playwright E2E, Lighthouse CI and Sentry; delivered in 50+ vertical slices on Vercel.

Python AI Engineer (Intern) — Allianz Global Investors **Apr 2024 – Sep 2024**

Paris · GenAI Hub: central platform bringing generative AI services to business teams

- Built the backend of a centralized GenAI platform: Python/Flask REST APIs, microservices architecture with CQRS (Clean Architecture), role- and permission-based access control.
- Built RAG and content-processing pipelines on Azure OpenAI and Azure AI Search, orchestrated with Semantic Kernel, delivering **4 GenAI services**: conversational assistant, financial report generation, multilingual translation, summarization.
- Took the prototype toward production with Docker, Kubernetes, Terraform and Datadog monitoring; built the UIs in React, TypeScript and Redux Toolkit.
- Presented technical choices and performance indicators to stakeholders, leading to **budget approval for the production phase**.

Software Engineer (Intern) — We Beautiful Minds **Apr 2021 – Sep 2021**

Casablanca · SaaS connecting influencers with their followers

- Built REST APIs in Java Spring Boot and the React/TypeScript front end of a SaaS platform; delivered the MVP and the development roadmap, contributing to product design (Figma) and Scrum ceremonies.

PROJECTS

OpenCRQ — cyber risk quantification platform (FAIR) | *FastAPI, NumPy, pgmpy, React, TypeScript* opencrq.fly.dev

- Built a vectorised Monte Carlo engine in NumPy: 100,000 simulated years × 4 scenarios in ~489 ms, reporting expected annual loss, VaR95/VaR99 and loss exceedance curves; profiled it to cut runtime by 11.5% and memory by 24% with bit-identical results.
- Designed a Bayesian network built from MITRE ATT&CK mappings and a provenance graph tracing every parameter to its source data; LLM explanations with no model arithmetic; 89% test coverage, strict mypy, 8 ADRs.

Clause — AI contract risk reviewer | *Next.js 16, TypeScript, OpenAI GPT-4o, Zod*

- Built a six-stage LLM pipeline (extract → classify → segment → retrieve → analyze → score) streaming results as NDJSON, with RAG over a 34-pattern risk knowledge base using precomputed embeddings and cosine similarity.
- Made LLM output reliable: strict JSON schemas validated with Zod, parallel batches with retry, and verbatim checks so every quoted span exists in the source document.

Chess Terminal — open-source desktop chess trainer | *Electron, React, TypeScript* github.com/medbm5/Chess-Terminal

- Shipped a cross-platform desktop app (~14K lines) running the Stockfish engine as WebAssembly in Web Workers, with SM-2 spaced repetition, Glicko-2 ratings and a streaming import of ~30,000 Lichess puzzles.
- Hardened Electron (context isolation, local-first IndexedDB); CI/CD in GitHub Actions with type-checking, 28 test files, content validation and automated installers for Windows, macOS and Linux.

MateToScroll — chess-powered screen-time blocker (Android) | *React Native, Expo, Kotlin, TypeScript*

- Wrote a custom Kotlin Expo native module using an Android Accessibility Service to block selected apps until time is earned back by solving puzzles; 30K puzzles in a bundled 8 MB SQLite file, 90% coverage gate on core logic.

SEC Filing Analyst — traceable LLM agent pipeline | *Python, FastAPI, LangGraph, PostgreSQL/pgvector*

- Built an agentic pipeline with typed stages and resumable state in Postgres, a verification step rejecting any claim not tied to its source, and a 151-question evaluation measuring automatic vs. human grading agreement (Cohen's kappa).

Also: self-healing ML serving platform (Kubernetes, Argo Rollouts canary, PSI drift detection, MLflow) · flight-delay lakehouse on Databricks (PySpark, Delta Lake, Auto Loader, 30M+ flights).

EDUCATION

MSc Machine Learning for Data Science — Université Paris Cité **2023 – 2024**

Engineering Degree (Master's level) — Innovation & IT Project Management, INPT Rabat **2019 – 2022**

Preparatory Classes MPSI (Mathematics & Computer Science) — CPGE Reda Slaoui **2017 – 2019**

CERTIFICATIONS

AWS Certified: Solutions Architect – Associate, Developer – Associate, Machine Learning – Associate, AI Practitioner, Cloud Practitioner · Microsoft Azure AI Engineer Associate (in preparation)

LANGUAGES

Arabic (native) · French (fluent) · English (fluent)